

Exam Review Questions

The following is a small selection of questions that can help you prepare for the final exam. This is not a complete exam review sheet. You will be required to consult various reference sheets for data to help solve a few of the problems.

Structure & Properties Unit

1. What is the letter code for a subshell with $\ell = 3$?
2. How many orbitals are found in the f subshell?
3. The ground state electron configuration of a phosphorus atom is $1s^2 2s^2 2p^6 3s^2 3p^3$. More specifically, $3p^3$ can be expanded to: $3p_x 3p_y 3p_z$. What does this mean?
4. What feature of their ground state electron configurations distinguishes the transition elements from other elements?
5. Explain the term isoelectronic. Provide an example.
6. Explain the term polar bond.

7. What is the maximum number of electrons allowed within the $n = 4$ region of space?
What is the maximum number of electrons belonging to the principal quantum number $n = 4$?
8. Write the longhand ground state electron configuration for boron & write the short-hand ground state electron configuration for sulphur.
9. Draw energy level diagrams to show the ground state electron configurations of argon & zinc.
10. How many electrons are in the valence energy level of bromine & xenon?
11. Name the neutral element that has the following ground state electron configuration:
 $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$.
12. What is a resonance structure? Draw all resonance structures for the carbonate ion, CO_3^{2-} .

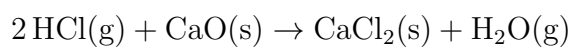
13. Explain why NH_3 is trigonal pyramidal whereas BCl_3 is trigonal planar.
14. Using VSEPR theory, explain why the bond angle in CO_2 is 180° .
15. Draw the Lewis structures for AsCl_5 & BrO_3^- . Name the shape of these structures. Indicate if the molecules are polar & the type of intermolecular forces that would result between 2 molecules of AsCl_5 & between 2 molecules of BrO_3^- .
16. Explain why gases may be compressed while solids & liquids may not be compressed to any extent.
17. What is coordinate covalent bonding?
18. For the element titanium, Ti:
- a) Predict the most stable electron configuration for neutral titanium.

- b) Without looking at the periodic table for the actual charges, predict the 2 most stable ionic charges for titanium. Which one should be more stable?
- c) On the periodic table, it says Ti^{3+} is stable, but is not the most stable ionic form. Provide a possible electron configuration that would explain why Ti^{3+} is considered a stable ionic charge.
19. Draw & explain the valence bonding & hybridization that occurs in ethane, ethene & ethyne.
20. Which will have the higher boiling point?
- a) SF_2 or SBr_2
- b) Cl_2 or I_2
- c) Cu or C_3H_8
21. What kind of crystal solid is formed from MgO ?

Energy Changes & Rates of Reactions Unit

1. How much energy is released when 15.0 grams of water vapour at 112.0°C becomes −15.0°C solid ice?

2. Calculate ΔH° for the following reaction:



using enthalpies of formation values.

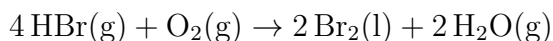
3. For the reaction: $3\text{H}_2 + \text{N}_2 \rightleftharpoons 2\text{NH}_3$

- a) How does the rate of disappearance of hydrogen compare to the rate of disappearance of nitrogen?
- b) How does the rate of appearance of ammonia compare to the rate of disappearance of nitrogen?
- c) If the rate of the appearance of NH_3 is $2.5 \times 10^{-6} \text{ mol}/(\text{L} \cdot \text{s})$, how fast are H_2 & N_2 being used?

4. The rate law for the reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ is $r = k[\text{NO}]^2[\text{O}_2]$. At 25°C , $k = 7.1 \times 10^9 \text{ L}^2/(\text{mol}^2 \cdot \text{s})$.
- What is the rate of reaction when $[\text{NO}] = 0.0010 \text{ mol/L}$ & $[\text{O}_2] = 0.034 \text{ mol/L}$?
 - Is the reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ an elementary step? Explain why.
5. Write the equation for the formation of the explosive TNT (trinitrotoluene), $\text{C}_7\text{H}_5\text{O}_6\text{N}_3(\text{s})$, from its elements at standard state.
6. When 5.60 kJ of thermal energy is added to 167 grams of gaseous ammonia, NH_3 , at 45.0°C , the temperature of the gas rises to 60.0°C . From this data determine the following:
- the specific heat capacity of NH_3
 - the molar heat capacity of NH_3
7. From the following three thermochemical equations, calculate the enthalpy change for the reaction: $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
- | | |
|---|-----------------------------------|
| $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ | $\Delta H^\circ = -25 \text{ kJ}$ |
| $3\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{Fe}_3\text{O}_4 + \text{CO}_2$ | $\Delta H^\circ = -47 \text{ kJ}$ |
| $\text{Fe}_3\text{O}_4 + \text{CO} \rightarrow 3\text{FeO} + \text{CO}_2$ | $\Delta H^\circ = +38 \text{ kJ}$ |

8. A copper container has a mass of 305 grams & contains 255 grams of water. When 1.01 grams of propanol, $\text{C}_3\text{H}_8\text{O}$, is burned in the container, the copper & contents increase in temperature by 28.6°C . Calculate the molar enthalpy of combustion of propanol using this experimental data.
9. A 50.0 gram piece of aluminum is heated to 100°C & then put into a beaker containing 150 mL of water at 20.0°C . Assuming no loss of thermal energy to the surroundings, calculate the final temperature of the water. Describe what is happening on the molecular level.
10. A 4.05 gram sample of sulphur is burned in excess oxygen. The energy produced raises the temperature of 1.00 L of water from 23.15°C to 32.03°C . From this information calculate ΔH_f° of sulphur dioxide.

11. The initial rate of production of bromine was measured using different initial concentrations of each reactant at 25.0°C. The equation & results are summarized below:



Initial Rate of Reaction (mol/(L·s))	Initial [HBr] (mol/L)	Initial [O ₂] (mol/L)
0.0042	0.010	0.010
0.0084	0.010	0.020
0.0126	0.030	0.010

- a) Determine the relationship between rate of decomposition & the concentration of each reactant.
- b) Determine the rate law for the reaction & the overall order of the reaction.
- c) Determine the value of the rate constant, k . Include units.
- d) Calculate ΔH° for the reaction as written using ΔH_f° values. Is the reaction endothermic or exothermic?
- e) Draw the potential energy diagram for the reaction. Label all critical points & values.
- f) What would happen if the reaction occurred at 30.0°C? Explain why.

Chemical Systems & Equilibrium Unit

1. For the following reaction: $\text{Br}_2(\text{l}) + 5 \text{F}_2(\text{g}) \rightleftharpoons 2 \text{BrF}_5(\text{g})$, $K_{eq} = 1.0 \times 10^{18}$.
 - a) Write the equilibrium law expression for the reaction.
 - b) At equilibrium, is the reaction mostly reactants or products dominant?
 - c) What should you do to the equation if you want to apply the “100” rule assumption?
 - d) If 0.12 mol/L of BrF_5 is initially present, calculate the equilibrium concentrations of all compounds.
 - e) If the reaction is: $2 \text{Br}_2(\text{l}) + 10 \text{F}_2(\text{g}) \rightleftharpoons 4 \text{BrF}_5(\text{g})$, what is the value of K_{eq} ?
2. At 1000°C methane reacts with water as follows: $\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3 \text{H}_2$, all compounds are gaseous. In one experiment, the equilibrium concentrations of the gases were $[\text{CH}_4] = 2.97 \times 10^{-3} \text{ mol/L}$, $[\text{H}_2\text{O}] = 7.94 \times 10^{-3} \text{ mol/L}$, $[\text{CO}] = 5.45 \times 10^{-3} \text{ mol/L}$, & $[\text{H}_2] = 2.1 \times 10^{-3} \text{ mol/L}$. Calculate K_{eq} at this temperature.
3. For the following process: $2 \text{FeBr}_3(\text{s}) \rightleftharpoons 2 \text{FeBr}_2(\text{s}) + \text{Br}_2(\text{g})$, the equilibrium constant, K_{eq} , is 0.98. A 3.0 L reaction vessel contains 1.2 mol of FeBr_3 , 3.6 mol of FeBr_2 & 2.1 mol of Br_2 at this temperature. Is the system at equilibrium?

4. The equilibrium constant for the gaseous system $\text{SO}_3 + \text{NO} \rightleftharpoons \text{NO}_2 + \text{SO}_2$ was found to be 0.250 at a certain temperature. If 0.300 mol of SO_3 & 0.300 mol of NO were placed in a 2.00 L container & allowed to react, what would be the equilibrium concentrations of each gas?
5. The gaseous system $2\text{HCl} + \text{I}_2 \rightleftharpoons 2\text{HI} + \text{Cl}_2$ has $K_{eq} = 1.6 \times 10^{-34}$ at 25°C . Suppose 1.00 mol of HCl & 1.00 mol of I_2 was placed in a 1.00 L container. What would be the equilibrium concentrations of HI & Cl_2 in the container?
6. A salt whose formula is of the form MX has $K_{sp} = 2.0 \times 10^{-10}$. Another slightly soluble salt, MX_3 must have what K_{sp} value if the molar solubilities of the two salts are identical?
7. Draw the concentration vs time graph illustrating the following situations for $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$; $\Delta H = +179 \text{ kJ}$
- a) initially start with $[\text{NO}]$ only
 - b) increase the partial pressure of nitrogen gas at constant volume

- c) decrease the volume of the container
- d) add He(g) at constant pressure
- e) increase the temperature of the system

8. What are the ion concentrations in a saturated solution of silver sulphate, Ag_2SO_4 ?

9. For the insoluble calcium fluoride, CaF_2 , if the solubility is 0.125 g/100 mL at 20°C :

a) Calculate the K_{sp} value for CaF_2 at 20°C .

b) Will a precipitate of CaF_2 form when 0.084 grams of sodium fluoride, NaF , is dissolved in 1.00 L of a 0.010 mol/L aqueous solution of calcium chloride, CaCl_2 ?

10. Calculate the maximum iodide ion concentration that can be added to 0.010 mol/L in lead (II) nitrate, $\text{Pb}(\text{NO}_3)_2$ before precipitation occurs.

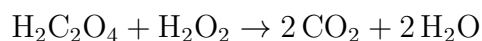
11. A 1.50 L solution of 0.250 mol/L aqueous sodium hydroxide, NaOH, is added to 1.00 L solution of 0.1875 mol/L magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$. Magnesium hydroxide, $\text{Mg}(\text{OH})_2$, precipitates from the mixture. Calculate the:
- a) maximum amount of $\text{Mg}(\text{OH})_2$ (in mol) that can form (assuming it cannot dissolve at all).
 - b) mass of magnesium hydroxide that precipitates.
 - c) pH for the solution.
12. Aspirin, $\text{HC}_9\text{H}_7\text{O}_4$, is a monoprotic acid whose $K_a = 3.27 \times 10^{-4}$. Does a solution of the copper (II) salt of aspirin in water test acidic, basic or neutral? In other words, is $\text{Cu}(\text{C}_9\text{H}_7\text{O}_4)_2$ acidic, basic or neutral?
13. Let's assume ascorbic acid is monoprotic. If 1.76 grams of ascorbic acid, $\text{C}_6\text{H}_8\text{O}_6$, was placed in 100.0 mL water. What is the pH of the solution?
14. An aqueous solution containing 0.302 grams of a triprotic acid, H_3X , requires

39.30 mL of a 0.120 mol/L solution of sodium hydroxide, NaOH, for the neutralization of all three protons. Calculate the molecular mass of the acid.

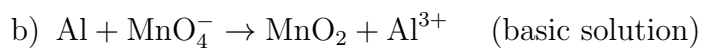
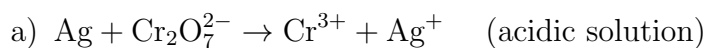
15. In an experiment, 0.030 L of 0.25 mol/L ethanoic acid CH_3COOH , was added to 0.020 L of 0.23 mol/L sodium hydroxide, NaOH. Find the pH of the resultant solution. *Hint: the base will neutralize only part of the acid. What is the pH of the leftover acid?*
16. What is the pH of a buffer solution that contains 0.10 mol/L NaHCO_3 & 0.12 mol/L H_2CO_3 ?

Electrochemistry Unit

1. What is the difference between oxidation numbers & valence charges?
2. What are the oxidation numbers in the following: sulphur in Cu_2SO_4 & carbon in $\text{C}_2\text{H}_3\text{O}_2$?
3. Identify the reactant that is oxidized & the reactant that is reduced in each reaction. Which reactant is the oxidizing agent?



4. Balance the following redox reactions:



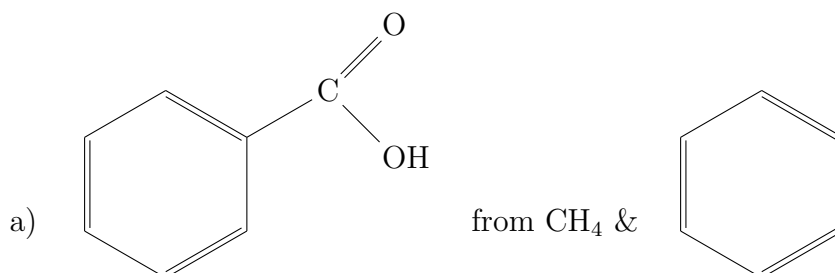
5. How can a solution of aqueous metal cations be isolated & purified in a metal plating process?

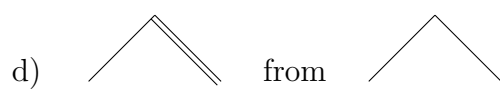
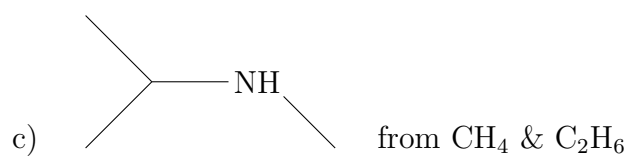
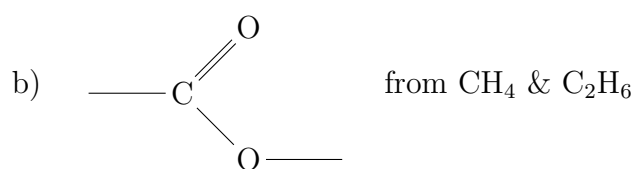
6. Which of the following is the best oxidizing agent in aqueous solution: Cl_2 , Cu^{2+} or Ag^+ ?
7. Determine if the following unbalanced reaction is spontaneous or not: $\text{Cd}^{2+} + \text{Fe} \rightarrow \text{Cd} + \text{Fe}^{2+}$.
8. Calculate the minimum cell voltage of a lithium iodide, LiI , electrolytic cell, where iodine must oxidize.
9. Draw & fully label the galvanic cell with the following notation: $\text{Al(s)} \mid \text{Al}^{3+}(\text{aq}) \parallel \text{Cu}^{2+}(\text{aq}) \mid \text{Cu(s)}$. Include appropriate chemicals, as needed. Include the direction of electron & ion flow. Calculate ΔE .
10. An aqueous solution of silver nitrate, AgNO_3 , is subjected to electrolysis for 30.0 minutes using a constant current of 1.50 amperes. Calculate the mass of silver metal deposited at the cathode.

11. Calculate the standard cell potential (ΔE) at 25°C for the reaction: $\text{Fe}^{2+}(\text{aq}) + \text{Ag}^+(\text{aq}) \rightleftharpoons \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$.
12. Aluminum metal is produced by the electrolysis of molten aluminum oxide, Al_2O_3 . Consider one such cell with a potential difference between the anode & cathode of 5.00 V. Electrons are supplied at a rate of 1.00×10^5 amperes. How much time (in hours) will it take for one cell to produce 1100 kg of aluminum?
13. Write the spontaneous reaction that occurs when $\text{Ni}(\text{ClO}_3)_2$ is mixed with CrCl_2 in the presence of NaOH .

Organic Chemistry Unit

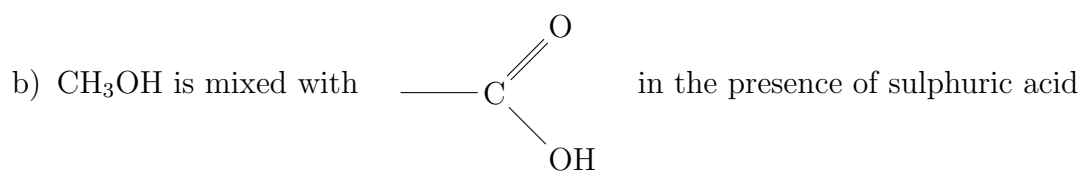
1. Which of alkanes or alkynes (assuming both have the same number of carbon atoms) have the higher boiling point & why?
2. Which of ethers or alcohols (assuming both have the same number of carbon atoms) have the higher solubility in oil & why?
3. Draw all the different structural isomers of C_5H_8 .
4. What is the difference between a structural & a geometric isomer?
5. Synthesize the following compounds (using the shortest number of steps) from the indicated starting materials. Use appropriate reagents in your synthesis.

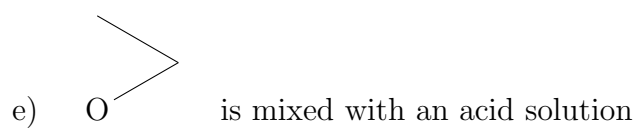
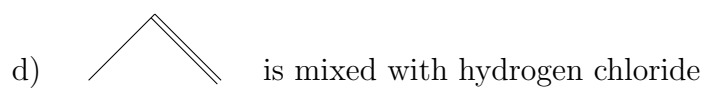
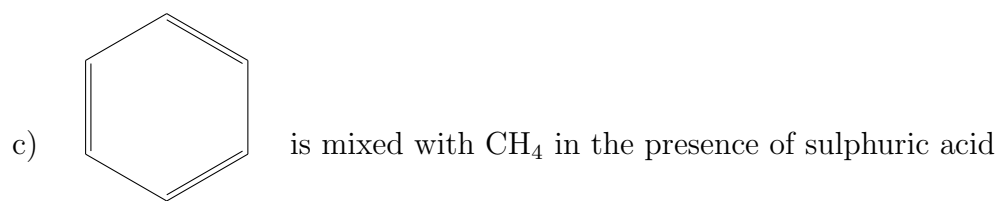




6. Predict the products for the following reactions. If no reaction occurs, write NR.

a) CH_4 is mixed with chlorine gas in the presence of UV light





7. What is a polymer? Provide an example of a condensation & addition polymer.